

THE 2-DOMINATION NUMBER AND THE UPPER MEDIAN DEGREE: A PROOF OF GRAFFITI.PC CONJECTURE 387

JUN QING

ABSTRACT. Let G be a nonempty finite simple graph of order n , and let $m(G)$ be the upper median of its degree sequence. We prove that

$$\gamma_2(G) \leq n - m(G) + 1,$$

where $\gamma_2(G)$ is the minimum cardinality of a set $D \subseteq V(G)$ such that every vertex outside D has at least two neighbors in D . This confirms Graffiti.pc Conjecture 387. In fact, the argument shows that the connectedness hypothesis in the original formulation is unnecessary. The proof uses the complement graph and a minimally linearly dependent family of polynomials encoding selected non-neighborhoods.

1. INTRODUCTION

All graphs in this note are finite and simple. A set $D \subseteq V(G)$ is *2-dominating* if every vertex in $V(G) \setminus D$ has at least two neighbors in D . The minimum cardinality of such a set is the *2-domination number* $\gamma_2(G)$. For general background on domination in graphs, see [2].

Let the degree sequence of an n -vertex graph G be written in nondecreasing order as

$$d_1 \leq d_2 \leq \cdots \leq d_n.$$

Its *upper median degree* is

$$m(G) = d_{\lfloor n/2 \rfloor + 1}.$$

A conjecture generated by Graffiti.pc and recorded by DeLaViña, Larson, Pepper, and Waller [1, Conjecture 1] predicts, for connected graphs, the upper bound

$$(1) \quad \gamma_2(G) \leq n - m(G) + 1.$$

The displayed statement of that conjecture in [1] contains the opposite inequality sign; however, the sentence immediately preceding it explicitly states the upper bound (1), and the subsequent discussion and partial results use the upper-bound direction. Thus the displayed sign is a typographical error.

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We prove the conjectured inequality for every nonempty finite simple graph, without assuming connectedness.

Theorem 1. *Let G be a nonempty finite simple graph of order n . Then*

$$\gamma_2(G) \leq n - m(G) + 1.$$

The only algebraic ingredient is the following elementary lemma about a minimal linear dependence among polynomials.

2. A POLYNOMIAL LEMMA

Lemma 1. *Let B be a finite set, let $t \geq 0$, and choose pairwise distinct real numbers a_z for $z \in B$. Suppose $A_1, \dots, A_r$ are t -element subsets of B and*

$$p_i(X) = \prod_{z \in A_i} (X - a_z) \quad (1 \leq i \leq r).$$

Assume that the indexed family $(p_i)_{i=1}^r$ is minimally linearly dependent over $\mathbb{R}$. Put

$$P = \bigcap_{i=1}^r A_i, \quad s = |P|.$$

Then

- (1) $2 \leq r \leq t + 2$;
- (2) *every element of $B \setminus P$ belongs to at most $r - 2$ of the sets $A_1, \dots, A_r$;*
- (3) $s \leq t + 2 - r$.

Proof. Each p_i is nonzero, so $r \geq 2$. Minimal dependence implies that any $r - 1$ of the polynomials are linearly independent. Since all p_i lie in the $(t + 1)$ -dimensional vector space of real polynomials of degree at most t , we obtain $r - 1 \leq t + 1$, and hence $r \leq t + 2$.

Choose a nontrivial dependence

$$(2) \quad \sum_{i=1}^r c_i p_i = 0.$$

Minimality implies $c_i \neq 0$ for every i . If some $z \in B$ belonged to exactly $r - 1$ of the sets A_i , then evaluating (2) at $X = a_z$ would leave exactly one nonzero summand. Indeed, if $z \notin A_j$, then

$$p_j(a_z) = \prod_{u \in A_j} (a_z - a_u) \neq 0$$

because the numbers a_u are pairwise distinct. This contradiction shows that an element occurring in at least $r - 1$ of the sets must occur in all r of them. Consequently, every element outside P occurs in at most $r - 2$ sets.

Finally, define

$$g(X) = \prod_{z \in P} (X - a_z)$$

and write $p_i = g\tilde{p}_i$. Multiplication by the nonzero polynomial g is injective, so the indexed family $(\tilde{p}_i)_{i=1}^r$ is again minimally linearly dependent. Each $\tilde{p}_i$ has degree $t - s$. Therefore any $r - 1$ of them are linearly independent in a vector space of dimension $t - s + 1$, giving

$$r - 1 \leq t - s + 1.$$

Equivalently, $s \leq t + 2 - r$. $\square$

3. PROOF OF THE MAIN THEOREM

Proof of Theorem 1. Write $m = m(G)$. If $m = 0$, then $V(G)$ is a 2-dominating set and

$$\gamma_2(G) \leq n < n + 1 = n - m + 1.$$

Hence assume $m \geq 1$. Let $F = \overline{G}$ and put

$$t = n - 1 - m, \quad q = t + 2 = n - m + 1.$$

In particular, $0 \leq t \leq n - 2$ and $2 \leq q \leq n$.

Define

$$H = \{x \in V(G) : d_F(x) \leq t\}, \quad B = V(G) \setminus H,$$

and write $h = |H|$ and $k = |B|$. Since

$$d_F(x) = n - 1 - d_G(x),$$

a vertex lies in H exactly when its degree in G is at least m . By the definition of the upper median, at least $\lceil n/2 \rceil$ vertices have degree at least m . Thus

$$(3) \quad h \geq \left\lceil \frac{n}{2} \right\rceil, \quad k \leq \left\lfloor \frac{n}{2} \right\rfloor, \quad h \geq k.$$

First suppose that $k \leq t + 1$. Since $k < q \leq n$, choose a q -element set D with $B \subseteq D$. Every vertex $v \notin D$ belongs to H , so

$$d_F(v, D) \leq d_F(v) \leq t.$$

As $v \notin D$, its neighbors in F among the vertices of D are exactly its nonneighbors in G among those vertices. Hence

$$d_G(v, D) = |D| - d_F(v, D) \geq q - t = 2.$$

Thus D is 2-dominating.

It remains to consider the case

$$(4) \quad k \geq t + 2.$$

For each $x \in H$, we have

$$|N_F(x) \cap B| \leq d_F(x) \leq t.$$

Using (4), extend $N_F(x) \cap B$ to a t -element subset $A_x \subseteq B$. Choose pairwise distinct real numbers a_z for $z \in B$, and define

$$p_x(X) = \prod_{z \in A_x} (X - a_z) \quad (x \in H),$$

where the empty product is 1 when $t = 0$.

The vector space of real polynomials of degree at most t has dimension $t + 1$. By (3) and (4),

$$h \geq k \geq t + 2,$$

so the indexed family $(p_x)_{x \in H}$ is linearly dependent. Choose an inclusion-minimal dependent subfamily, indexed by a set $C \subseteq H$, and put $r = |C|$. Apply Lemma 1 to the sets $(A_x)_{x \in C}$. With

$$P = \bigcap_{x \in C} A_x, \quad s = |P|,$$

the lemma gives

$$(5) \quad s \leq t + 2 - r,$$

and every element of $B \setminus P$ belongs to at most $r - 2$ of the sets A_x with $x \in C$.

Because $2 \leq r \leq t + 2$, we have $0 \leq t + 2 - r \leq t$. Together with (5) and $|B| = k \geq t + 2$, this allows us to choose $Y \subseteq B$ such that

$$P \subseteq Y, \quad |Y| = t + 2 - r.$$

Set

$$D = C \cup Y.$$

Since $C \subseteq H$ and $Y \subseteq B$, the union is disjoint and

$$|D| = r + (t + 2 - r) = t + 2 = q.$$

Let $v \notin D$. If $v \in H$, then simply

$$d_F(v, D) \leq d_F(v) \leq t.$$

Now suppose $v \in B$. Since $P \subseteq Y$ and $v \notin Y$, we have $v \notin P$. Lemma 1 therefore shows that v belongs to at most $r - 2$ of the sets A_x , $x \in C$. As $N_F(x) \cap B \subseteq A_x$ for every $x \in C$, and F is undirected,

$$d_F(v, C) = |\{x \in C : v \in N_F(x) \cap B\}| \leq r - 2.$$

Consequently,

$$d_F(v, D) = d_F(v, C) + d_F(v, Y) \leq (r - 2) + |Y| = t.$$

We have shown that every $v \notin D$ satisfies $d_F(v, D) \leq t$. Therefore

$$d_G(v, D) = |D| - d_F(v, D) \geq (t + 2) - t = 2.$$

Thus D is a 2-dominating set of cardinality

$$|D| = q = n - m + 1,$$

and the theorem follows. $\square$

Remark 1. The bound is sharp for every $n \geq 2$: for the complete graph K_n , one has $m(K_n) = n - 1$ and $\gamma_2(K_n) = 2$.

Remark 2. Connectedness is not used anywhere in the proof. Hence the conclusion strictly strengthens the original formulation of Graffiti.pc Conjecture 387.

REFERENCES

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